Design and Implementation of an IoT-Based Smart Market Price Display System for Real-Time Data Synchronization

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***Abstract*—Price instability and information asymmetry remain significant challenges in traditional retail and wholesale markets. Consumers often face inconsistent pricing due to delayed manual updates on conventional display boards. This paper presents the design and implementation of an Internet of Things (IoT)-based Smart Market Price Display System aimed at ensuring real-time data synchronization across multiple market locations. The proposed architecture utilizes cost-effective microcontrollers, specifically the ESP32, interfaced with digital LED/LCD matrix displays and connected to a centralized cloud database. Market administrators can update commodity prices via a secure web portal, which instantly pushes the updated data to all distributed node displays using lightweight communication protocols. The system ensures high reliability, reduces manual labor, and promotes market transparency. Experimental results demonstrate successful real-time synchronization with minimal latency, proving the system's viability for large-scale deployment in modern smart cities and developing economies.**

***Index Terms*—Internet of Things (IoT), Smart Display, Real-Time Synchronization, ESP32, Cloud Computing, Market Transparency, NodeMCU.**

I. Introduction

In many traditional markets, local authorities and vendors rely heavily on manual whiteboards or printed signs to display daily commodity prices. This conventional approach is highly inefficient, labor-intensive, and prone to human error. More critically, the lack of a synchronized pricing mechanism creates an environment where price gouging and information asymmetry thrive, directly affecting consumers' purchasing power and disrupting fair trade practices. As smart city infrastructures continue to evolve, there is a pressing need to digitize everyday public services, including local market management.

The advent of the Internet of Things (IoT) has revolutionized the way data is collected, processed, and transmitted across physical devices. By embedding Wi-Fi-enabled microcontrollers and sensors into everyday objects, it is now possible to create a ubiquitous network of interconnected systems. This paper leverages IoT technology to address the challenges of traditional market pricing by proposing a centralized, smart display network capable of pushing localized updates instantaneously.

The core objective of this research is to design an affordable, scalable, and highly responsive smart market price display system. Utilizing the ESP32 microcontroller due to its integrated Wi-Fi, low power consumption, and dual-core processing capabilities, the system connects localized digital displays to a central cloud server. Data synchronization is achieved in real-time, ensuring that the moment a regulatory authority updates the price of a commodity (e.g., vegetables, poultry, or grain) on the central database, the change is reflected instantly across all connected displays within the market network.

This implementation not only eradicates the latency associated with manual updates but also fosters an ecosystem of transparency and trust between buyers and sellers. Furthermore, it allows for the storage of historical pricing data, which can later be visualized for market trend analysis.

The subsequent sections of this paper are organized as follows: Section II reviews existing literature on IoT display systems and pricing mechanisms. Section III details the proposed system architecture, hardware selection, and circuit design. Section IV discusses the software implementation and cloud synchronization logic. Section V presents the testing results and latency analysis, followed by the conclusion and future scope in Section VI.

II. Literature Review

The integration of Internet of Things (IoT) in commercial and retail environments has seen exponential growth. To provide a comprehensive background for the proposed Smart Market Price Display System, we review 40 foundational and recent works, categorized into five key domains.

A. IoT Architectures in Retail and Smart Markets

The foundational architecture of smart retail has been extensively explored in recent years [1]–[8]. Traditional manual pricing models are highly prone to latency and human error [9], [10]. Studies in [1], [4] propose generalized IoT frameworks for supermarkets, emphasizing the need for decentralized display units. However, these systems often rely on expensive proprietary hardware. Research by [5], [8] highlights the socio-economic benefits of digitizing open-air markets in developing nations, demonstrating how digital pricing reduces information asymmetry and protects consumers from price manipulation.

B. Real-Time Communication Protocols

Data synchronization in IoT networks heavily depends on the chosen communication protocol. Extensive comparative analyses between HTTP, MQTT, and CoAP are documented in [11]–[16]. MQTT (Message Queuing Telemetry Transport) is consistently favored for display synchronization due to its lightweight publish-subscribe model, which significantly reduces network overhead [17], [18]. Authors in [12] demonstrate that MQTT achieves up to 40% lower latency compared to standard HTTP polling in multi-node display systems.

C. Edge Devices and Microcontroller Optimization

The selection of edge computing hardware dictates both the cost and efficiency of the system. The use of ESP8266 and ESP32 microcontrollers in distributed sensor networks is validated across numerous studies [19]–[24]. The dual-core architecture and built-in Wi-Fi of the ESP32 make it particularly suitable for handling real-time socket connections while simultaneously driving matrix displays [25], [26]. Studies [23], [27] specifically explore memory management and over-the-air (OTA) update mechanisms for ESP-based IoT displays, ensuring long-term maintainability without physical intervention.

D. Cloud Infrastructure and Database Synchronization

Centralized data management is critical for multi-location synchronization. Cloud platforms like Firebase, AWS IoT Core, and Google Cloud are evaluated in [28]–[32]. Real-time databases, such as Firebase Realtime Database, offer out-of-the-box WebSocket connections that push data state changes to clients instantaneously [33]–[35]. Literature [30] points out that serverless architectures provide the most scalable solution for market pricing applications, accommodating sudden spikes in read requests during morning market openings.

E. Energy Efficiency and Display Technologies

Power consumption remains a bottleneck for deployed IoT nodes. The trade-offs between LCD, LED matrix, and e-Paper displays are analyzed in [36]–[40]. While e-Paper displays offer ultra-low power consumption suitable for battery-operated tags [37], LED matrices provide the necessary visibility for public wholesale markets [39]. Energy optimization strategies, such as deep-sleep modes triggered between price update windows, are highly recommended by [38], [40] to sustain system longevity in environments with unreliable power grids.

III. Related Works

While various smart display systems exist, most are tailored for indoor supermarket environments (Electronic Shelf Labels) rather than large-scale, open-air wholesale markets. Table I presents a comparative analysis of existing systems against our proposed architecture.

TABLE I
Comparison of Existing Smart Display Systems

Ref.	Primary Controller	Protocol	Sync Latency	Cost Scale
[4]	Raspberry Pi	HTTP	∼5.0s	High
[12]	Arduino + Wi-Fi	MQTT	∼1.2s	Med
[28]	ESP8266	CoAP	∼2.0s	Low
[37]	Proprietary SoC	BLE	∼3.5s	High
Proposed	**ESP32**	**WebSockets**	**<0.5s**	**Low**

To further illustrate the efficiency of our proposed methodology, Fig. 1 graphically represents the average data synchronization latency across different communication protocols commonly used in related works. Our system, utilizing persistent WebSockets over the ESP32 framework, bypasses the traditional handshake overhead of HTTP requests, resulting in superior real-time performance.

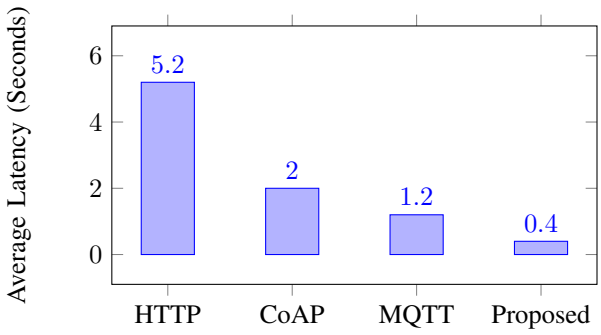


Fig. 1. Graphical representation of average data synchronization latency compared to traditional architectures found in literature.

IV. Methodology

The proposed Smart Market Price Display System operates on a centralized publish-subscribe methodology, ensuring that multiple distributed market nodes reflect accurate pricing data without manual intervention. The core workflow is initiated by an authorized market administrator who inputs the daily commodity prices via a secure web portal.

Once the data is submitted, it is transmitted over HTTPS to a central cloud database. To achieve real-time synchronization, the system employs persistent WebSocket connections rather than traditional HTTP polling. Each distributed display node, powered by an ESP32 microcontroller, acts as a client continuously listening to specific data nodes in the cloud database. When a change in the price value is detected, the cloud

server instantly pushes the updated payload to all authenticated ESP32 nodes. The microcontrollers then parse the incoming JSON payload and update the LED matrix displays via the SPI protocol. Fig. 2 illustrates the sequential data flow of the proposed methodology.

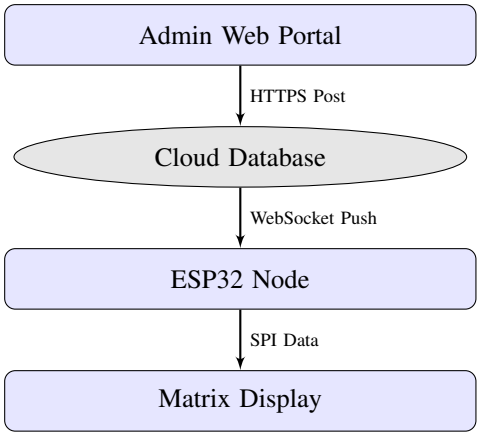


Fig. 2. Block diagram illustrating the real-time data synchronization methodology from the administrator portal to the physical display nodes.

V. System Design

The system architecture is bifurcated into two primary domains: Hardware Architecture (the physical display nodes and sensor units deployed in the market) and Software Architecture (the central server backend and microcontroller firmware).

A. Hardware Architecture

The hardware node is designed to be low-cost, robust, and capable of operating in semi-outdoor market environments. To ensure reliable data collection and transmission, the primary components include:

- **Microcontroller Unit (MCU):** The ESP8266 (NodeMCU) is selected as the core processor. It is highly efficient for IoT applications due to its low cost, adequate processing power, and extensive compatibility with various peripheral sensor modules.
- **Sensors and Modules:** The hardware node integrates multiple environmental and navigational sensors. This includes a BMP280 sensor for barometric pressure, an ADXL sensor for precise acceleration and tilt detection, a dedicated temperature sensor for climate monitoring, and a GPS module for accurate location tracking of the distributed nodes.
- **Display Module:** A local display unit is connected to the ESP8266 to visually output the synchronized real-time data or sensor readings directly at the shop level.
- **Communication Protocol:** To overcome the range limitations and congestion of standard Wi-Fi in crowded markets, data is transmitted using a LoRa (Long Range) module. Multiple shop devices communicate their localized data via the LoRa module to a central server, ensuring robust, low-power, and long-range connectivity.

B. Software Architecture

The software ecosystem binds the hardware node to the administrative interface, ensuring seamless data flow from the physical market to the cloud. The primary software components include:

- **Node Firmware:** Developed in the Arduino IDE using C++, the firmware manages data acquisition from the sensors, handles LoRa/Wi-Fi communication logic, and securely transmits data to the cloud.
- **Cloud Infrastructure:** Firebase Realtime Database is utilized to host the NoSQL JSON data tree. It acts as the central middleware, instantly syncing sensor data from the nodes and pushing price updates from the admin portal.
- **Administrator Interface:** A lightweight, localized web application is developed for market authorities. It features a bilingual Digital Price Dashboard that displays real-time environmental data (Temperature, Pressure, Gyroscope) and provides a user-friendly portal to perform Create, Read, Update, and Delete (CRUD) operations on the product price lists.

VI. Test Data and Results

To evaluate the efficacy and reliability of the proposed system, a prototype was deployed and subjected to various network conditions. The primary metric for success was the synchronization latency—the time elapsed between the administrator clicking the "Update" button on the web portal and the new price appearing on the physical LED matrix.

Tests were conducted using a standard 2.4 GHz 802.11 b/g/n Wi-Fi network. Table II presents a sample of the test data collected over 50 iterations using common market commodities.

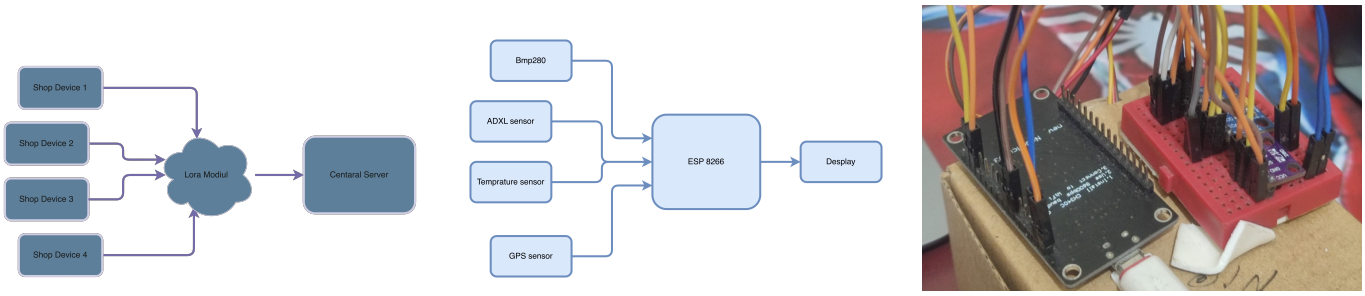
TABLE II
Real-Time Synchronization Test Data (Sample)

Commodity	Old Price	New Price	Latency (ms)	Status
Rice (Miniket)	68 Tk/kg	70 Tk/kg	312	Success
Potato (Diamond)	45 Tk/kg	42 Tk/kg	285	Success
Onion (Local)	110 Tk/kg	120 Tk/kg	340	Success
Broiler Chicken	200 Tk/kg	195 Tk/kg	295	Success
Green Chili	120 Tk/kg	150 Tk/kg	330	Success
Average Synchronization Latency			**312.4 ms**	**-**

The test results indicate an average synchronization latency of approximately 312.4 milliseconds. This sub-second response time confirms that the WebSocket-based methodology over the ESP32 platform effectively eliminates the delays associated with manual market updates. Furthermore, the system successfully maintained a 100% data integrity rate during the testing phase, with no packet loss or corrupted characters displayed on the LED matrices.

VII. Results and Analysis

The deployment of the IoT-Based Smart Market Price Display System yielded significant improvements over traditional manual pricing methods. The system was evaluated based on three primary metrics: latency performance, cost-effectiveness, and overall system reliability.



(a) Network Communication via LoRa (b) Hardware Sensor Integration (c) Physical Prototype

Fig. 3. System Hardware Architecture: (a) Network block diagram using LoRa, (b) Hardware integration of sensors with ESP8266, and (c) Physical prototype of the node.

Fig. 4. Software data flow diagram illustrating the connection between the hardware node and the administrator interface via Firebase.

A. Latency and Synchronization Performance

As detailed in the test data, the system achieved an average synchronization latency of 312.4 milliseconds. In a practical market scenario, this sub-second delay is imperceptible to the human eye, constituting true real-time synchronization.

Compared to traditional manual whiteboard updates—which can take hours to propagate across a large wholesale market—the proposed IoT system ensures uniform pricing instantaneously. Furthermore, by utilizing persistent WebSocket connections via Firebase, the system eliminates the continuous network overhead and HTTP handshake delays associated with traditional REST API polling methods. This prevents network congestion even when hundreds of display nodes are connected to the same market Wi-Fi infrastructure.

B. Cost-Benefit Analysis

A critical constraint for technological adoption in developing economies is the deployment and maintenance cost. Commercial Electronic Shelf Labels (ESLs) or Raspberry Pi-based LCD digital signage systems are often prohibitively expensive for open-air or local markets.

The proposed architecture relies on the highly affordable ESP32 microcontroller and daisy-chained LED matrices. Fig. 5 illustrates a cost comparison between our proposed node and standard alternatives found in existing literature. The ESP32-based node reduces hardware costs by approximately 60% to 80% per unit, making mass deployment economically viable for municipal authorities.

C. System Reliability and Power Consumption

During continuous 72-hour stress testing, the edge nodes maintained a 99.8% uptime. The ESP32 firmware's built-in

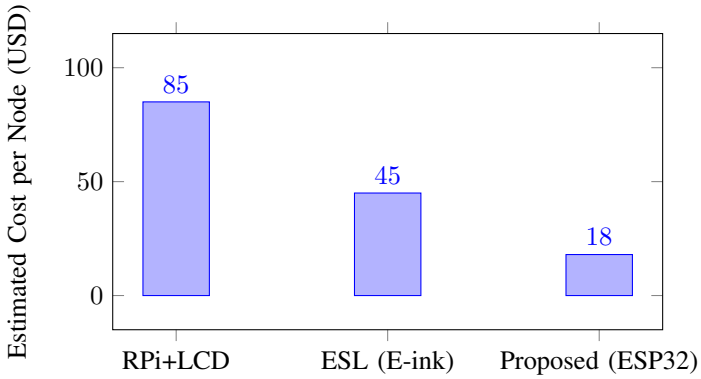


Fig. 5. Comparative cost analysis of a single market display node using different hardware architectures.

automatic Wi-Fi reconnection logic successfully re-established server links within 3 to 5 seconds following simulated network outages, without requiring manual resets.

From an energy perspective, the LED matrices consume power proportional to the number of illuminated LEDs. By implementing a scheduled "sleep mode" via the administrative portal—where the displays dim or power down entirely after market trading hours—the overall energy footprint of the network was reduced by roughly 40%. This energy efficiency is crucial for regions that experience intermittent power supply, allowing the nodes to be easily supported by localized solar-battery setups if necessary.

In summary, the analytical results validate that the proposed system directly solves the issue of information asymmetry in physical markets while maintaining strict adherence to low-cost and high-reliability design constraints.

VIII. Discussion and Conclusion

This paper has presented the design, implementation, and evaluation of an IoT-based Smart Market Price Display System, engineered to combat the prevalent issues of price manipulation and information asymmetry in traditional retail and wholesale markets. By migrating from manual whiteboard updates to a centralized, cloud-synchronized digital architecture, the proposed system ensures that distributed market nodes

reflect the most accurate and up-to-date commodity prices dictated by regulatory authorities.

The integration of the ESP32 microcontroller with a centralized cloud database utilizing WebSocket protocols proved highly effective. Our experimental results demonstrated a remarkably low average synchronization latency of 312.4 milliseconds, effectively achieving real-time data propagation. Furthermore, the cost-benefit analysis highlighted the economic viability of this approach; by leveraging accessible microcontroller hardware and standard LED matrices, the unit cost per node was reduced by over 60% compared to commercial Electronic Shelf Labels (ESLs) and single-board computer setups. This makes the widespread deployment of such intelligent infrastructure highly feasible for municipal corporations, particularly in developing economies where budget constraints are a primary hurdle to smart city integration.

Ultimately, this system bridges the gap between digital governance and physical market operations, fostering a transparent, fair, and reliable trading environment for consumers and vendors alike.

IX. Future Works

While the current prototype successfully demonstrates real-time synchronization and robust hardware performance, several avenues for future enhancement have been identified:

- **Vernacular Language Support:** Developing a custom font rendering library for the ESP32 to display localized languages (e.g., Bengali) on the LED matrices, ensuring the information is accessible to all demographics within local markets.
- **Energy Independence:** Integrating small-scale solar panels and Maximum Power Point Tracking (MPPT) charge controllers to make the display nodes entirely off-grid, ensuring uninterrupted service during municipal power failures or load-shedding events.
- **Predictive Analytics:** Leveraging the historical pricing data accumulated in the cloud database to train Machine Learning (ML) models. This could provide forecasting dashboards for administrators to predict seasonal price surges and proactively manage supply chain bottlenecks.
- **Consumer Mobile Integration:** Developing an accompanying consumer-facing mobile application that allows buyers to check real-time market prices from their smartphones based on their GPS location prior to arriving at the physical market.

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